# Fish Abundance Over Time

**Exploring common manipulations, working with dates and time, arrange(), and visualizing missing data**

## Set Up

```
# loading in packages
library(tidyverse) # general use
library(janitor) # cleaning data
library(here) # data organization
library(lubridate) # date and time
library(naniar) # to use gg_miss_var

# reading in data set
fish <- read.csv(here("data",
                      "annual_all_species_fish.csv"),
                 # changing -99999 values to be na
                 na = "-99999")
```

## Cleaning Data

```
# creating new object, fish_clean
fish_clean <- fish |>
  # cleaning the names
  clean_names() |>
  filter(group == "FISH") |>
  # renaming the date column to original_date
  rename(original_date = date) |>
  drop_na(dry_gm2) |>
  # using mutate and lubridate to convert original_date
  # into new date column that can be used to create time series
```

```
  mutate(
    date = ymd(original_date),
    start_of_year = floor_date(date, unit = "year")) |>
  # relocating columns to to left to be easier to look at
  relocate(start_of_year, dry_gm2, common_name, site)
```

### Common Manipulations: rename() & relocate()

#### Rename (function in dplyr)

The rename() function allows you to change the names of values for new ones.

In this data set: rename() was used to change the date column to be named original_date. This was useful so a different date column could be created through mutate, and date was the preferred column name.

#### Relocate (function in dplyr)

The relocate function allows for columns positioning in a data set to be changed. Relocate automatically pulls columns to the left, but can be changed using .before and .after.

In this data set: relocate() was utilized to pull columns created by mutate, which go to the very right side of data, to the left to be easier viewed along with other desired columns

### Working with Dates and Time

#### Utilizing lubridate package

Lubridate is a package that is used to help make working with dates and times easier.

In this data set:

- ymd() was used to turn the original dates into a version of dates that R can better recognize and put it in a year day month format
- floor_date() takes the new "date" column and rounds each date down to the first of the year This allows for the mean dry mass by year to later to explored in the time series.

### Other Functions Used

- clean_names(): used to simplify the column names from the original data file
- drop_na(): used to drop missing values from data to better prepare for summarize function use

## Exploring Data

```
# creating top_species_table
top_species_table <- fish_clean |>
  # grouping by species
  group_by(common_name) |>
  # summarize to make one row per species
  # sum to get the total dry biomass of each species
  summarize(total_dry_gm2 = sum(dry_gm2)) |>
  # ungrouping the data
  ungroup() |>
  # ranking from most to least abundant
  arrange(desc(total_dry_gm2)) |>
  # selecting top 5 species
  slice_head(n = 5)

# creating top_species_vector
top_species_vector <- top_species_table |>
  # using pull to extract column as a vector
  pull(common_name)
```

### Arrange (function in dpylr)

The arrange function allows for rows of a data frame to be ordered based on the values of specific columns, and can be done in either a descending or ascending order.

In this data set: arrange() is used to put the specified value, total_dry_gm2, in descending order. This was utilized to identify the top 5 abundant fish species from the data.

### Common Manipulations: pull()

### Pull (function in dpylr)

Pull extracts a single column from a data set and turns it into a plain vector.

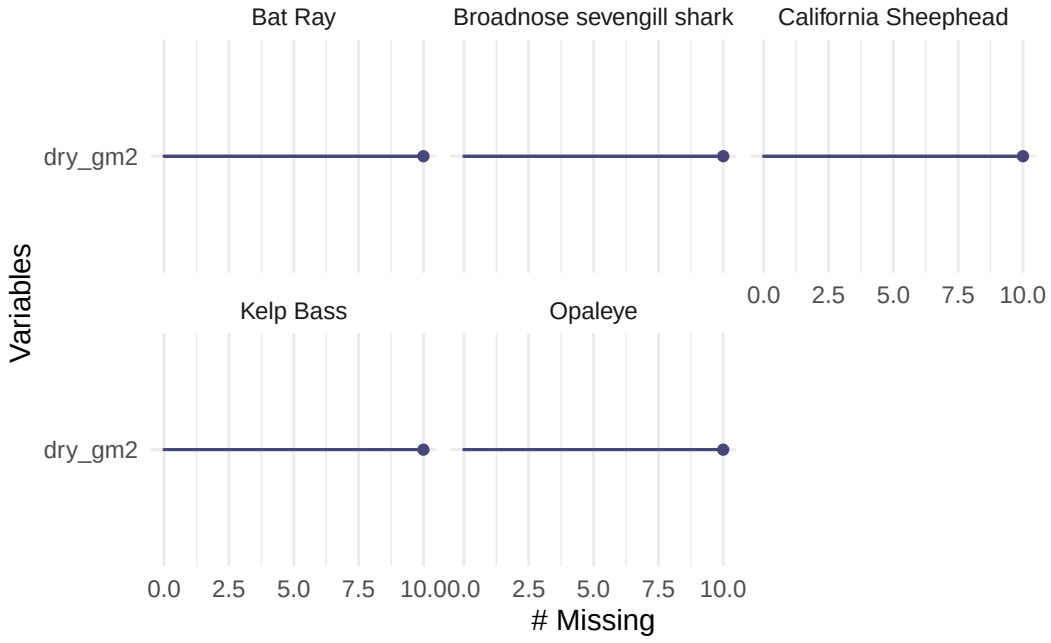
In this data set: pull() was used to extract the top species names from the top_species_table. This later allowed for this vector to be utilized when visualizing missing data and for preparing data for the time series.

### Other Functions Used

- group_by(): utilized to specify grouping by common_name for the summarize function
- summarize(): used to sum the dry_gm2 data for each species, and collapses data into one row per species
- ungroup(): undoes the grouping function
- slice_head(n=5): utilized to keep the first 5 rows, which is the 5 most abundant species

## Visualizing Missing Data

```
# looking at fish
fish |>
  clean_names() |>
  # filter for fish
  filter(group == "FISH") |>
  # filter for top species
  filter(common_name %in% top_species_vector) |>
  # selecting for dry mass
  select(dry_gm2, common_name) |>
  # using gg_miss_var to check for "na" data
  gg_miss_var(facet = common_name)
```



#### gg_miss_var (naniar package)

The function gg_miss_var allows you to visualize missing data from columns and does this through creating a ggplot showing the number of missing values for each variable selected.

In this data set: gg_miss_var() is used by selecting for the desired variables, dry_gm2 and common_name, and then faceting by common_name to show panels for each of the top_species_vector and how much dry_gm2 data is missing.

What this means: Each of the top 5 species are indicated to have 10 missing dry_gm2 data. This will be kept in mind through visualization and the following analysis.

### Preparing Data for Timeseries

```
# creating new object fish_ts
fish_ts <- fish_clean |>
  # filtering for the top_species_vector in common_name
  filter(common_name %in% top_species_vector) |>
  # using fct_relevel to change order within common_name
  mutate(common_name = fct_relevel(
    common_name,
    top_species_vector)) |>
  # grouping data by date and species
  group_by(start_of_year,
           common_name) |>
  # summarize to make one row per month
  # sum to get the mean dry mass for each species
  summarize(mean_dry_gm2 = mean(dry_gm2)) |>
  # ungrouping the data
  ungroup() |>
  # arranging by most abundant species and by date
  arrange(common_name, start_of_year)
```

### Top 5 Fish Species Time Series

```
# base plot
ggplot(fish_ts,
       # x-axis: time
       aes(x = start_of_year,
           # y-axis: abundance
```

```
          y = mean_dry_gm2,
          # coloring by species
          color = common_name)) +
# first layer: geom_line
geom_line() +
# second layer: geom_point
geom_point(size = 0.5) +
# faceting by species
facet_wrap(~ common_name,
           # putting into one column
           ncol = 1) +
scale_x_date(date_breaks = "3 years",
             date_labels = "%Y") +
labs(
  # adding a graph title
  title = "Top 5 Abundant Fish Through Time",
  # naming x axis Time
  x = "Time (years)",
  # naming y axis Total fish count
  y = "Mean dry biomass (g/m^2)") +
# customizing colors
scale_color_manual(values = c(
  "California Sheephead" = "deeppink1",
  "Bat Ray" = "slategray",
  "Kelp Bass" = "goldenrod2",
  "Opaleye" = "darkseagreen3",
  "Broadnose sevengill shark" = "dodgerblue4")) +
# adding custom theme
theme_minimal() +
# removing legend
theme(legend.position = "NONE")
```


Top 5 Abundant Fish Through Time
California Sheephead
Bat Ray
Kelp Bass
Opaleye
Broadnose sevengill shark
Mean dry biomass (g/m^2)
0
5
10
15
2001
2004
2007
2010
2013
2016
2019
2022
2025
Time (years)

This time series shows the mean dry biomass ($g/m^2$) from 2000 to 2026 for the five most abundant fish species: the California sheephead, bat ray, kelp bass, opaleye, and broadnose sevengill shark.

- **California sheephead:** stable populations at approximately 1-3$g/m^2$ through 2018 where a general increase is seen to approximately 7.5$g/m^2$ with fluctuations following while staying at elevated levels.
- **Bat ray:** highly variable with very large peaks in 2007 (approximately 17.5$g/m^2$) and in 2020 (approximately 10 $g/m^2$). Several years have 0$g/m^2$ also seen after spikes, with steady numbers at approximately 5$g/m^2$ starting in 2024
- **Kelp bass:** gradual increase from about 1 to 5$g/m^2$, with a dip in 2022 and 2025, followed by a high value in 2026 (10$g/m^2$)
- **Opaleye:** relatively stable populations between 0-4$g/m^2$ with an increase in 2026 (approximately 6$g/m^2$)
- **Broadnose sevengill shark:** zero dry biomass recorded from 2000-2019 with rising values in recent years and a notable peak in 2025 at 10$g/m^2$

These rises and falls in populations may be tied to ocean temperature shifts. Evaluating temperatures seen through time along with the top abundant species may bring ecological insight into how increasing ocean temperatures are impacting abundant species. Additionally, these changes may have a cascade effect through kelp forest food webs. For instance, tracking California sheephead, urchin, and kelp mean dry biomass ($g/m^2$) may bring insight to how top-down predation from California sheephead may impact these other species.